



Bottom Versus Charm Jet Tagging Using Machine Learning

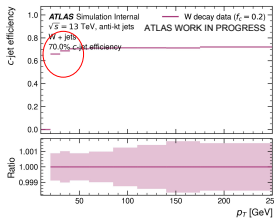
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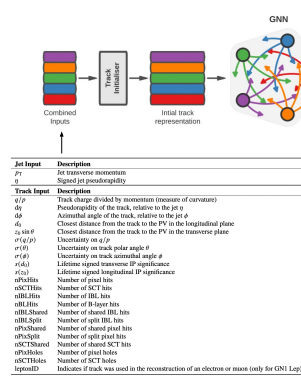
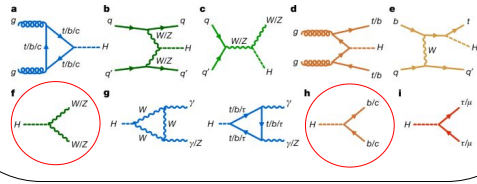
Introduction

Experimental Methods1

Conclusion and References

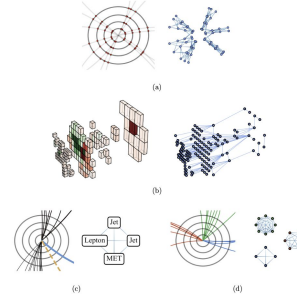


ATLAS has many well-developed flavor-tagging techniques for bottom quarks, but it **lags in performance for charms** with even worse performance at low p_T , which is a problem for many important new physics searches, primarily those concerning the Higgs Boson Decays



We use both jet and track information as model inputs, which can be well-represented in a graph format, and taking advantage of this, we pass this into our Graph Neural Network (GN2), from which we then classify calorimeter jets as bottom, charm or light

We have additional network tasks (auxiliary tasks) that help guide the network towards correct predictions



- Training on a sample that emphasizes low p_T gives modest but real **improvements in tagging performance**
- Classifying charm vs anti-charm appears doable and could have interesting applications

[1] J. Shlomi, P. Battaglia, and J.-R. Vlimant, arXiv preprint arXiv:2007.13681v2 (2020).

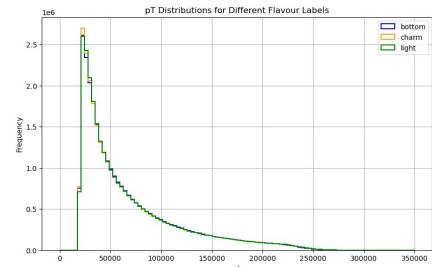
[2] ATLAS Collaboration. *Graph Neural Network Jet Flavour Tagging with the ATLAS Detector*. ATL-PHYS-PUB-2022-027, 1st June 2022.

[3] The ATLAS Collaboration. *Graph Neural Network Jet Flavour Tagging with the ATLAS Detector*. ATL-PHYS-PUB-2022-027, CERN, 1 June 2022.

[4] The ATLAS Collaboration. A detailed map of Higgs boson interactions by the ATLAS experiment ten years after the discovery. *Nature* 607, 52–59 (2022).

Results

Next Steps



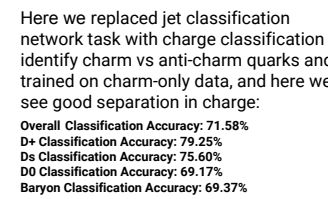
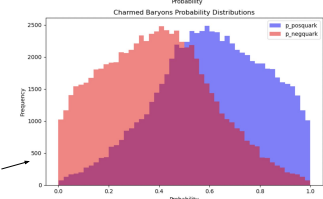
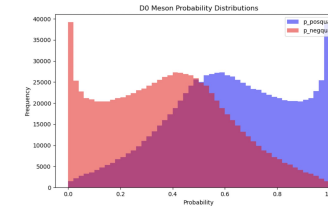
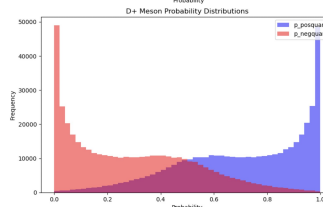
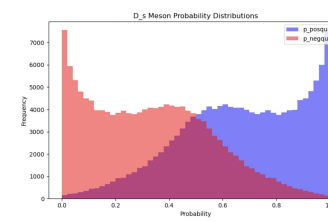
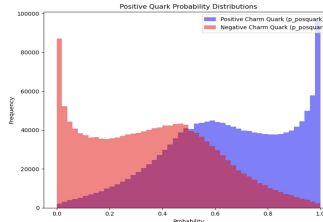
Retrained the GN2 Using MadGraph FFXF W + jets sample, which generates:

- W + b jets
 - W + c jets
 - W + light jets
- And from the plot we can see this is a heavily **low p_T sample**, which is where we want to improve performance

c jets → At 50% WP we have ~20% better b-jet rejection and ~10% worse light-jet rejection

b jets → At 50% WP we have ~40% better c-jet rejection and ~10% worse light-jet rejection

Baryon-Blindness for Quark Classification



Overall Classification Accuracy: 71.58%

D+ Classification Accuracy: 79.25%

Ds Classification Accuracy: 75.60%

D0 Classification Accuracy: 69.17%

Baryon Classification Accuracy: 69.37%

- **Refine** the charm vs anti-charm tagger for both its own task and to further improve GN2 Model performance
- More comprehensive studies of performance analysis
- Addition of more features:
 - Track Jets

PRELIMINARY GN2 RESULTS:

The ROC Curves show charge classification performance when it replaces the jet classification task and when it is an aux task, and how GN2 performance is impacted when it is used as an aux task. These are **preliminary** results which are positive indicators, but require further analysis.

